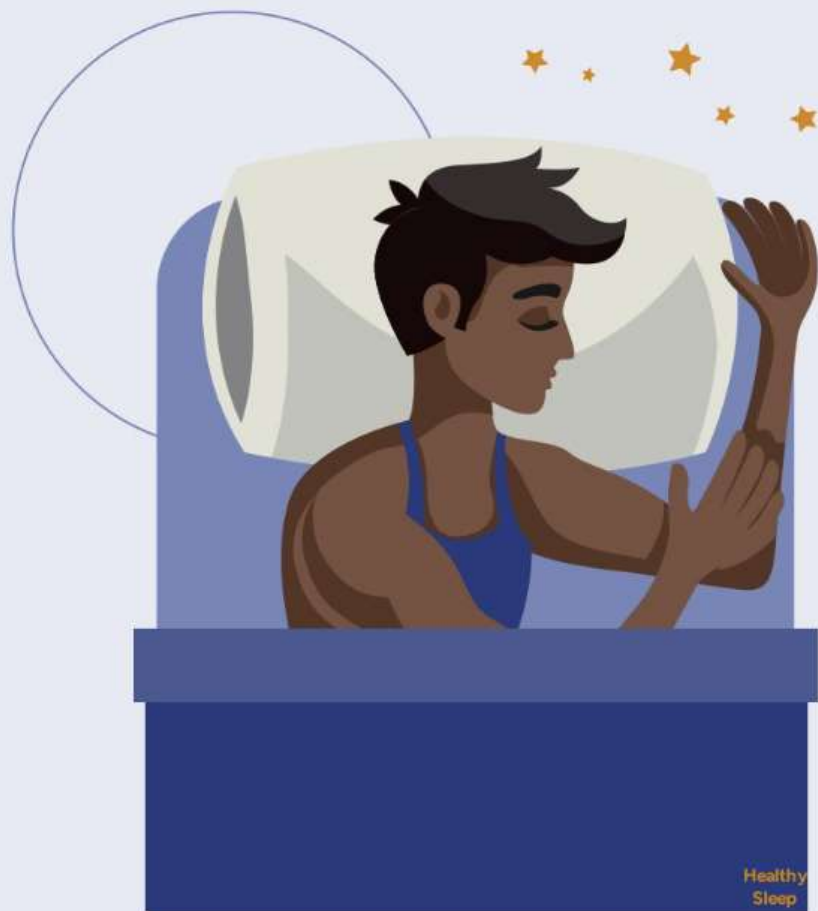




The SleepGuard

Smart Sleep Optimization Wristband

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Our team



Tom Huyghe

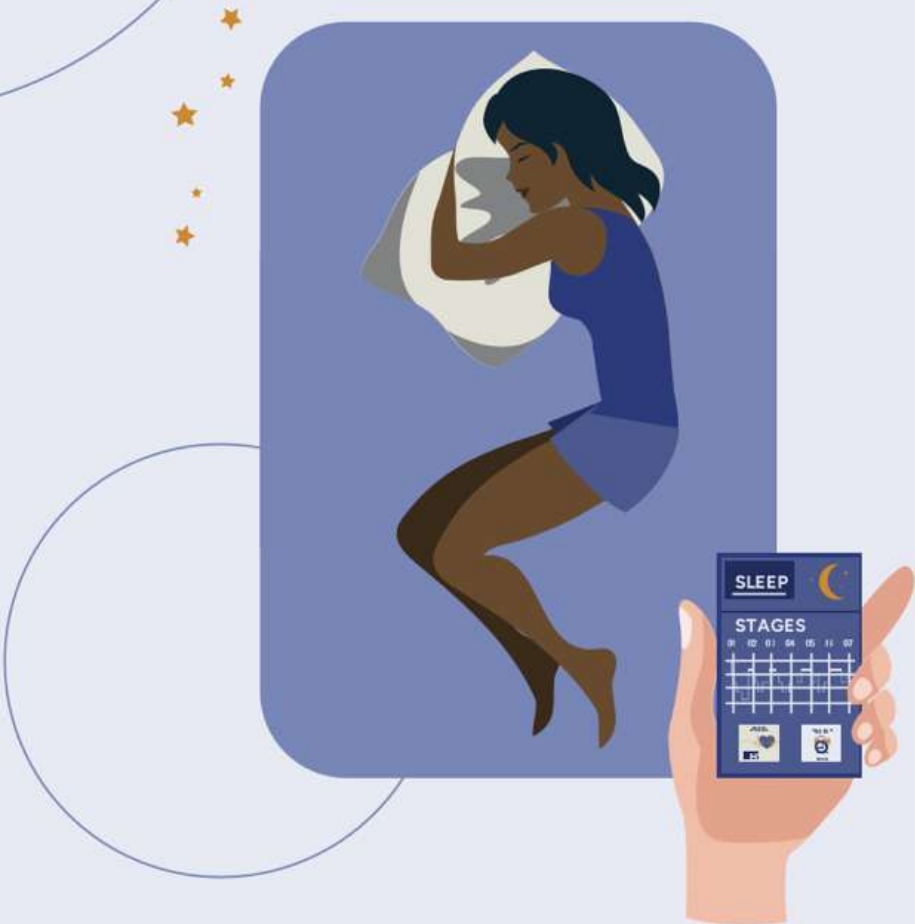


Adèle Faillé



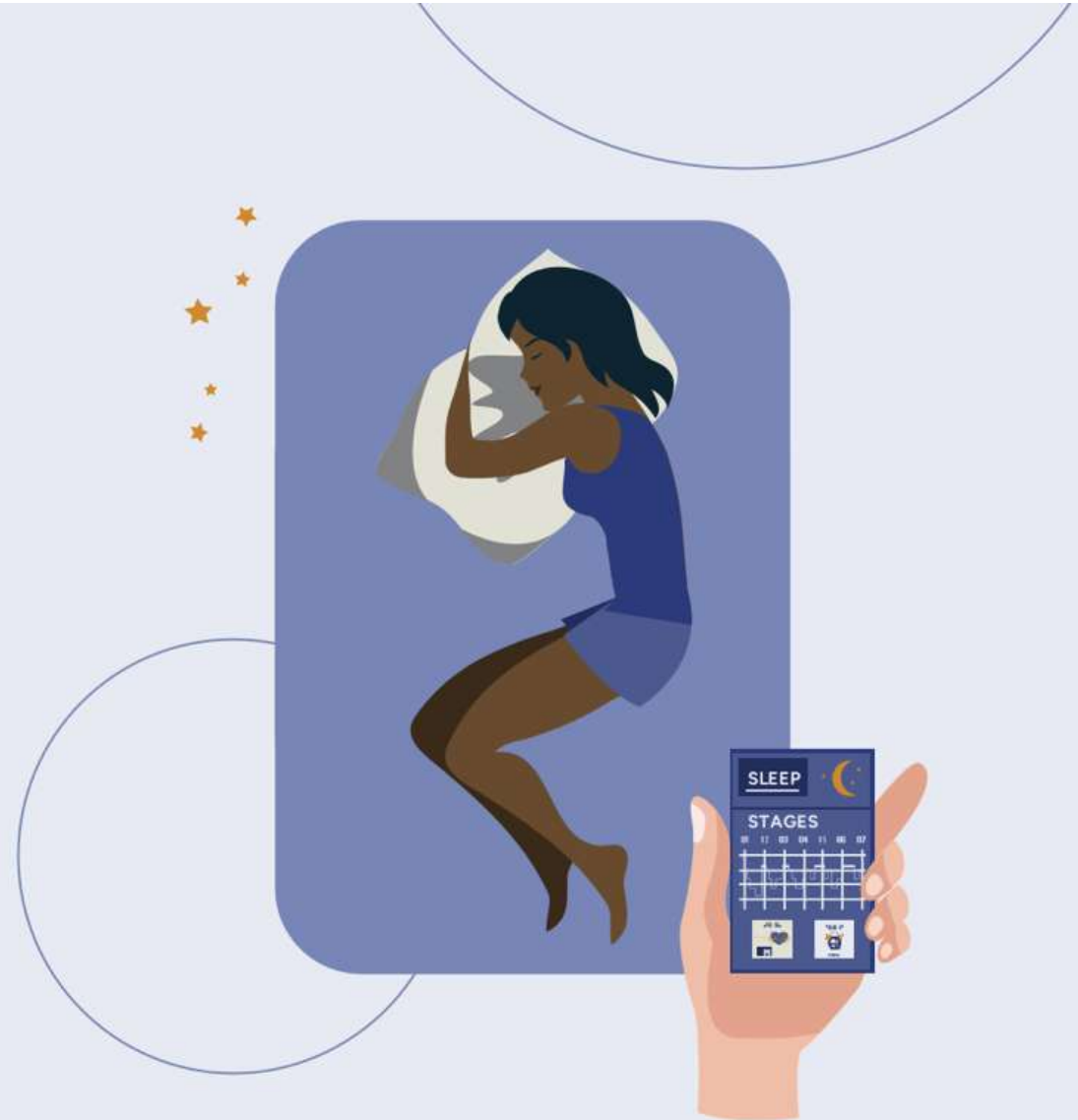
01

Abstract



Your sleep is precious!

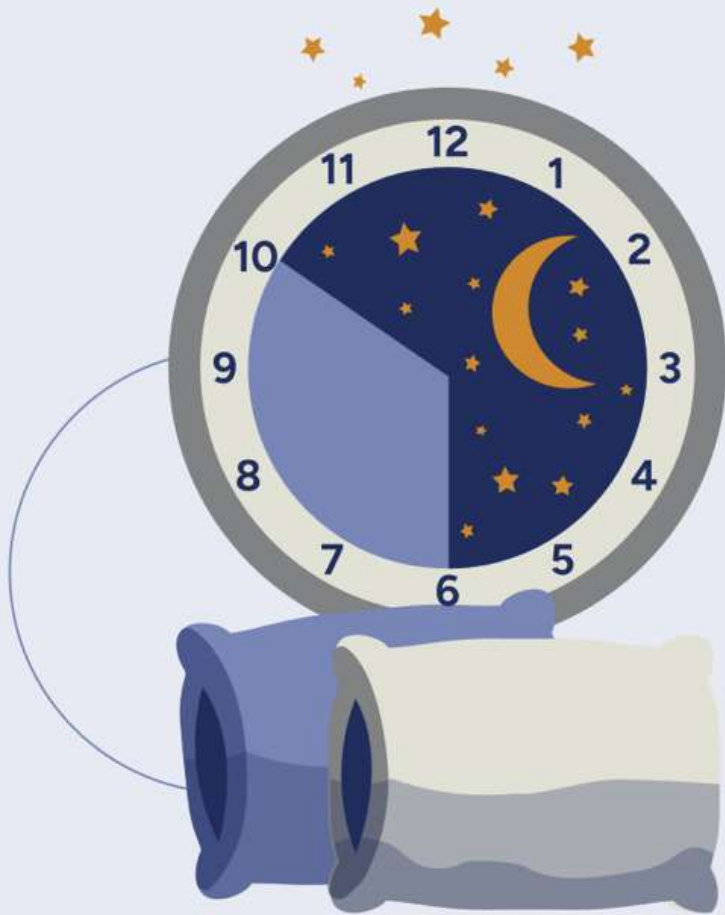
Introducing the SleepGuard to the world : why is this crucial ? How does it work ? How did we do it ? Investing in your sleep is investing in your overall health



02

Introduction





“The power of sleep. New research shows a good night’s rest isn’t a luxury, it’s critical for your brain and for your health.”

— Alice Park, PubMed, 2014

Introduction

The importance of monitoring your sleep for health and well-being.

- Renowned medical publications have featured numerous studies on the effects of sleep:
- Various technologies very helpful exist
 - Sleep tracking device
 - Sleep apps
 - Sleep aid devices
 - Sleep regulation technologies



03

Methodology





Objective

Sleep problems are prevalent worldwide. Many countries report an increase in cases of sleep disorders, partly due to modern lifestyles, stress, widespread screen use, and other factors.

Develop a wristband that uses DSP (Digital Signal Processing) techniques to monitor and improve the quality of a user's sleep by providing real-time feedback and recommendations.

How can we regulate sleep problems easily in our daily life ?

01 — Sleeping diary

02 — Sleep hygiene

03 — Exercise

04 — Light diner

05 — No Blue light

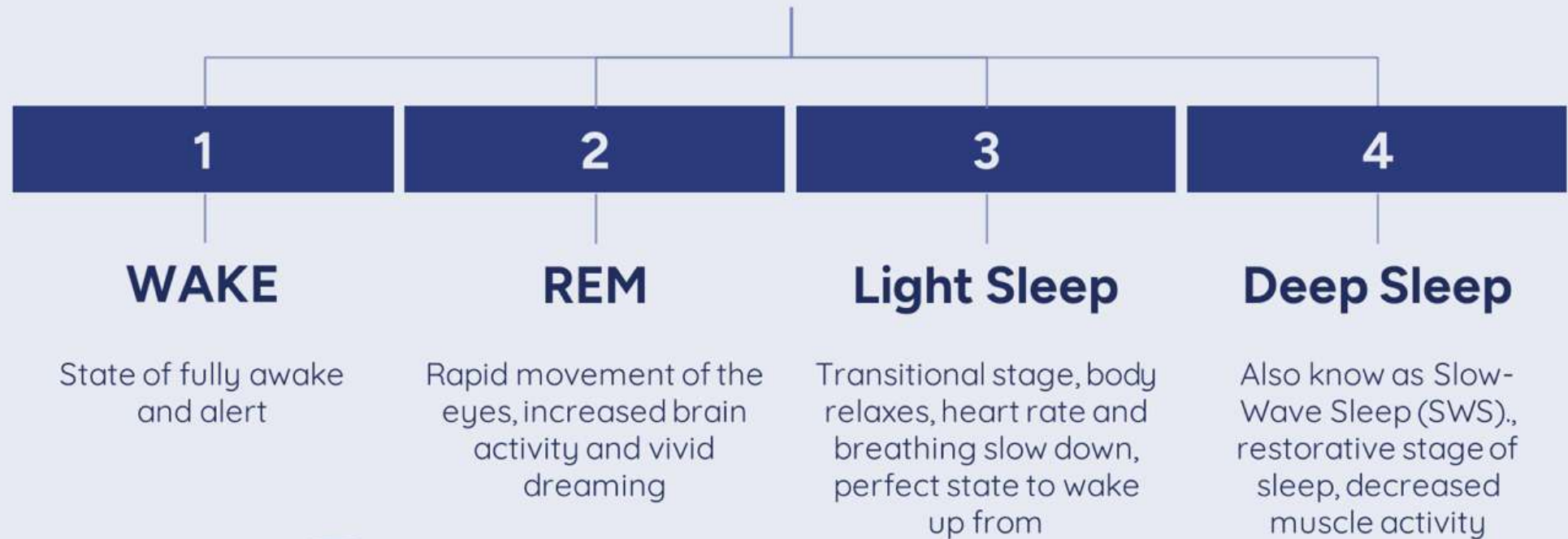
06 — Manage stress

07 — Limit naps

How does the sleep cycle work ?

Sleep cycle

Consists of several stages



90 minutes

A sleep cycle repeated 3 to 5 times in a night
depending on the person



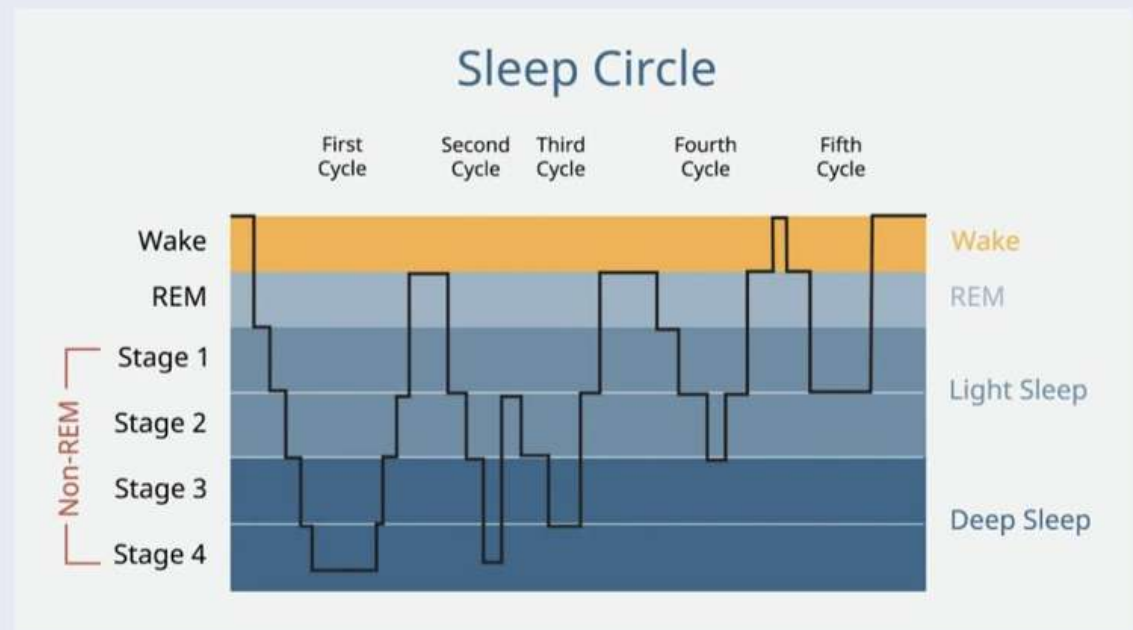
Average time spent in sleep stages

AWAKE	LIGHT SLEEP	DEEP SLEEP	REM SLEEP
2-5%	45-55%	13-23%	20-25%



"HRV hypnogram" (HRV stands for Heart Rate Variability). Represents the different sleep stages correlated with variations in heart rate throughout the night. It provides a temporal visualization of sleep phases in relation to changes in heart rate.

HRV hypnogram



Insomnia

One of the most common sleep problems worldwide

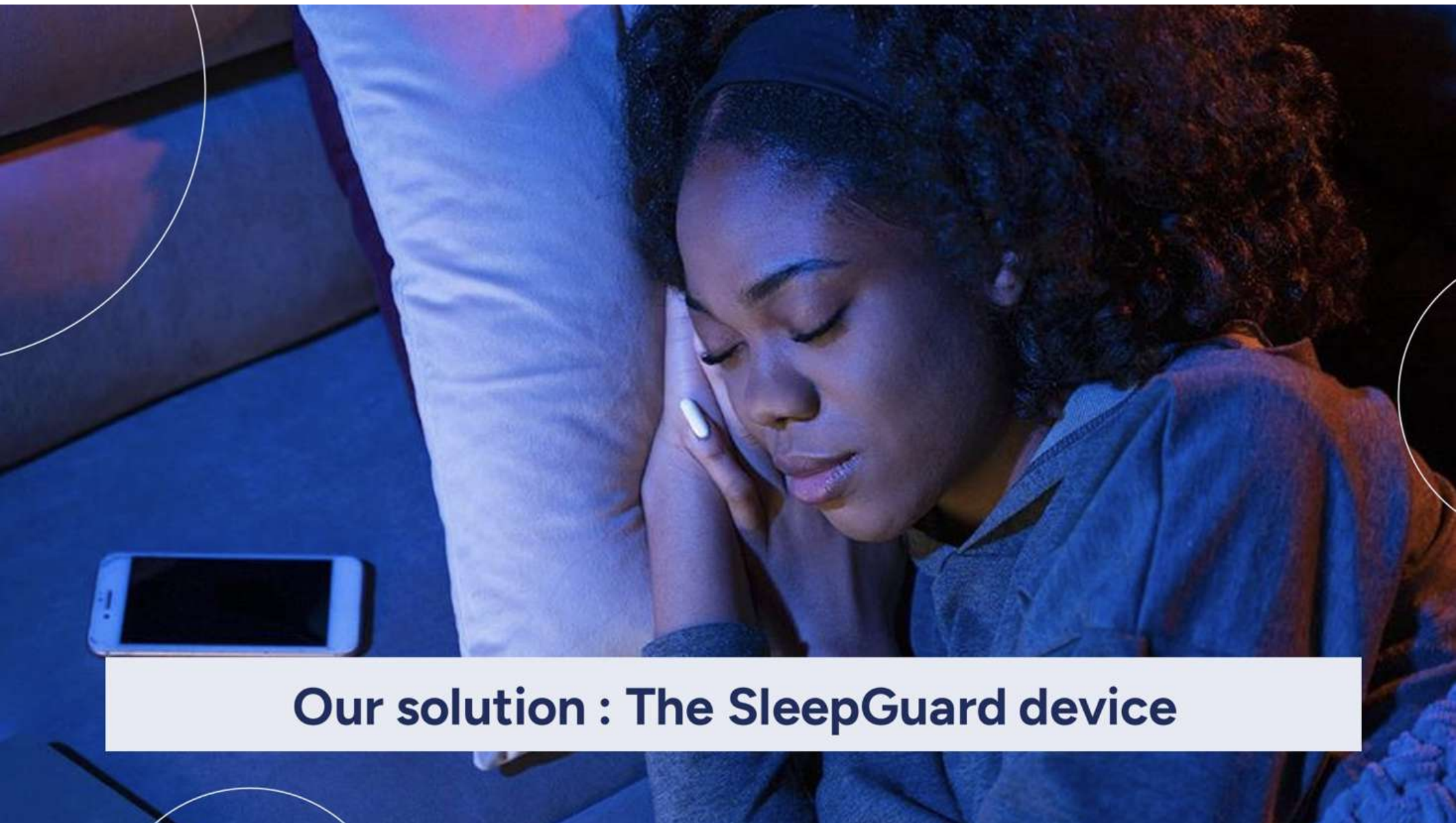
Sleep Apnea

Increasingly diagnosed and treated globally

Health Impact

On physical and mental health





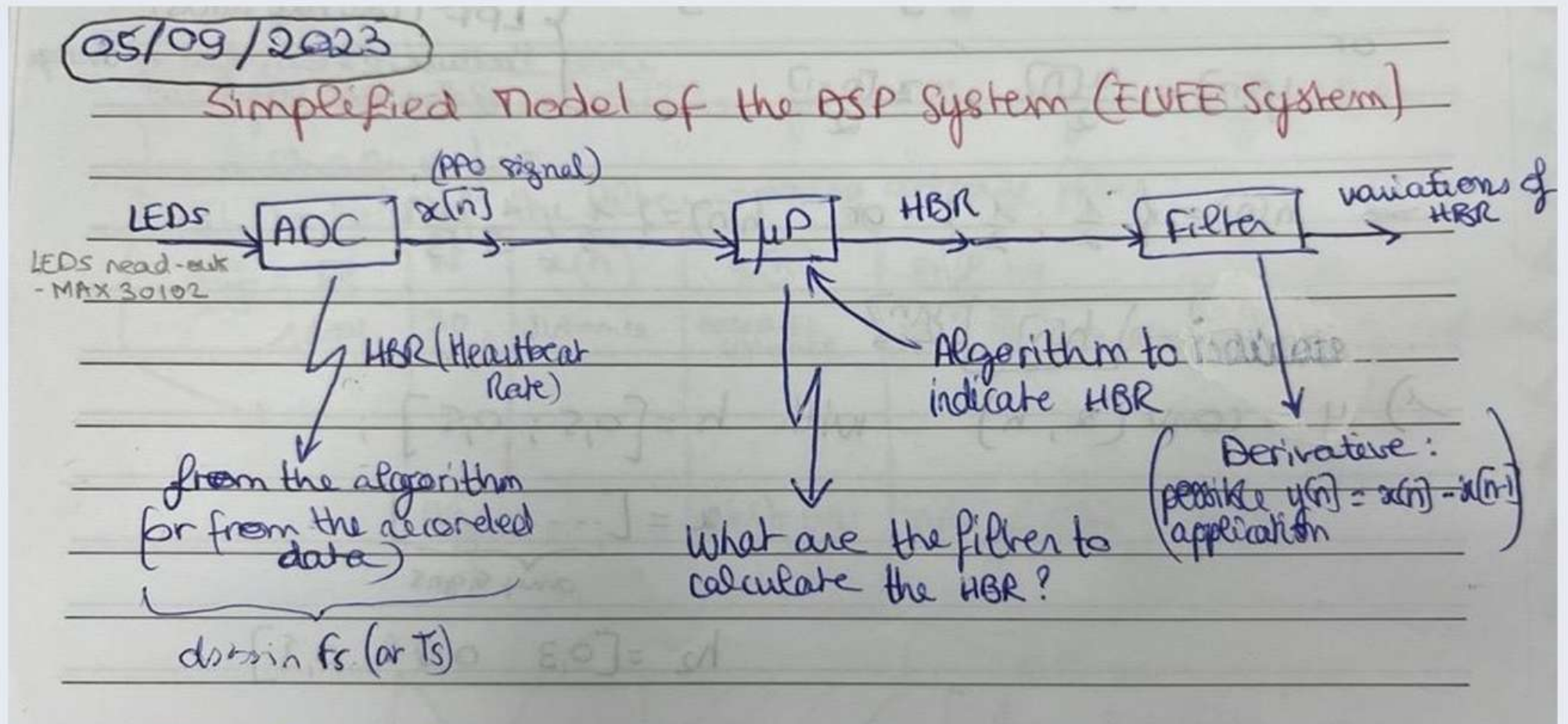
Our solution : The SleepGuard device

An application of the ELVEE Pulse

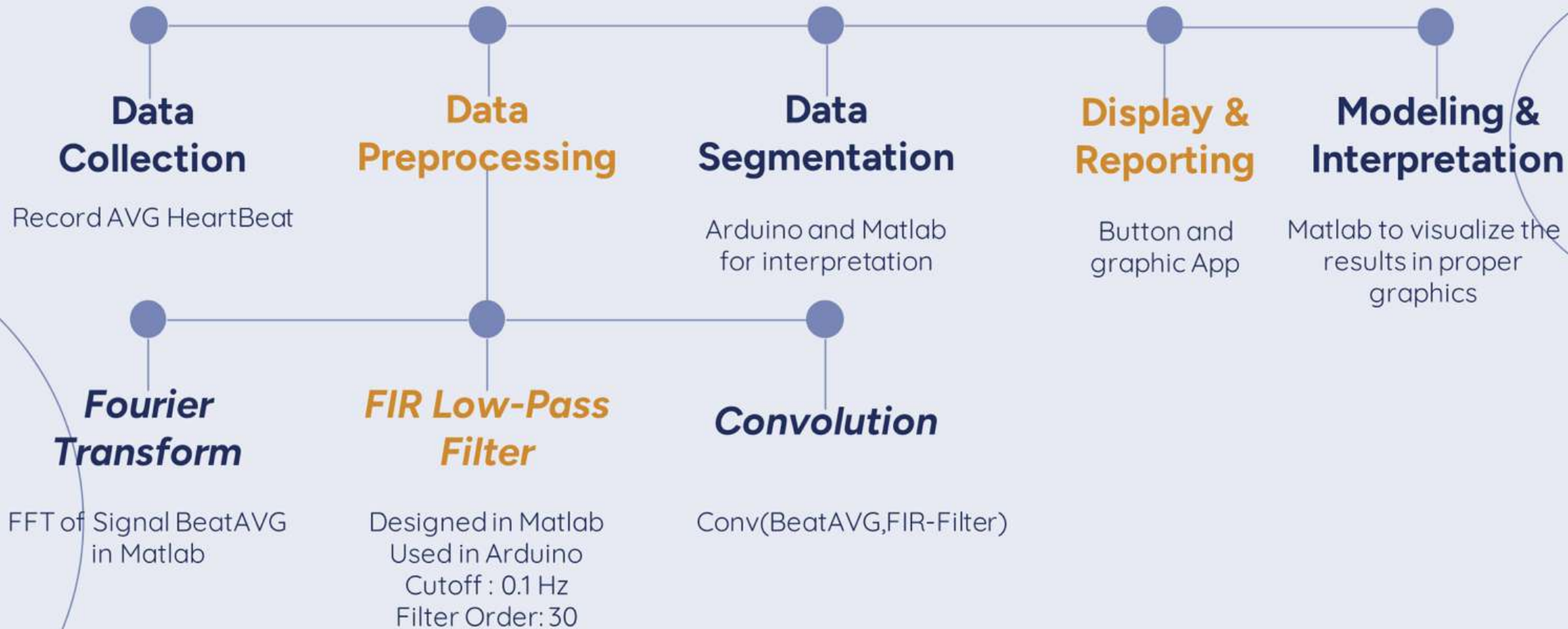
Routines to improve sleep

01	02	03
Monitoring	Analyzing	Adapting
1. Keep track of overall health	1. Obtain valuable data	1. Improving sleep habits
2. Offers insights into your sleep patterns	2. Identifying sleep disorders	

The plan



The Methodology we followed



Data Collection

TIME	BeatAvg
00:10:00	63.74
00:10:01	60.84
00:10:02	62.34
00:10:03	67.15
00:10:04	75.04
00:10:09	60.12
00:10:10	69.33
00:10:10	64.81
00:10:11	70.12
00:10:14	74.87
00:10:15	77.4
01:40:13	70.85
01:40:13	67.41
01:40:25	67.51
01:40:26	64.82
01:40:34	69.24
01:40:35	66.13
01:40:36	60.35
01:40:37	57.34
01:40:38	61.53
01:40:39	59.44
01:40:40	58.22
01:40:41	57.33

```
// Enregistre la valeur dans le fichier CSV
dataFile = SD.open("heart_rate_data.csv", FILE_WRITE);
if (dataFile)
{
    // Calcul du temps écoulé depuis le début de l'analyse
    unsigned long elapsedTime = millis() - startTime;
    int hours = elapsedTime / 3600000;
    int minutes = (elapsedTime % 3600000) / 60000;
    int seconds = (elapsedTime % 60000) / 1000;

    // Enregistre le temps au format HH:MM:SS et filteredBeatAvg dans le fichier CSV
    dataFile.print(hours);
    dataFile.print(":");
    dataFile.print(minutes);
    dataFile.print(":");
    dataFile.print(seconds);
    dataFile.print(",");
    dataFile.println(beatAvg);

    dataFile.close();
}
else
{
    Serial.println("Error opening file for writing");
}
```

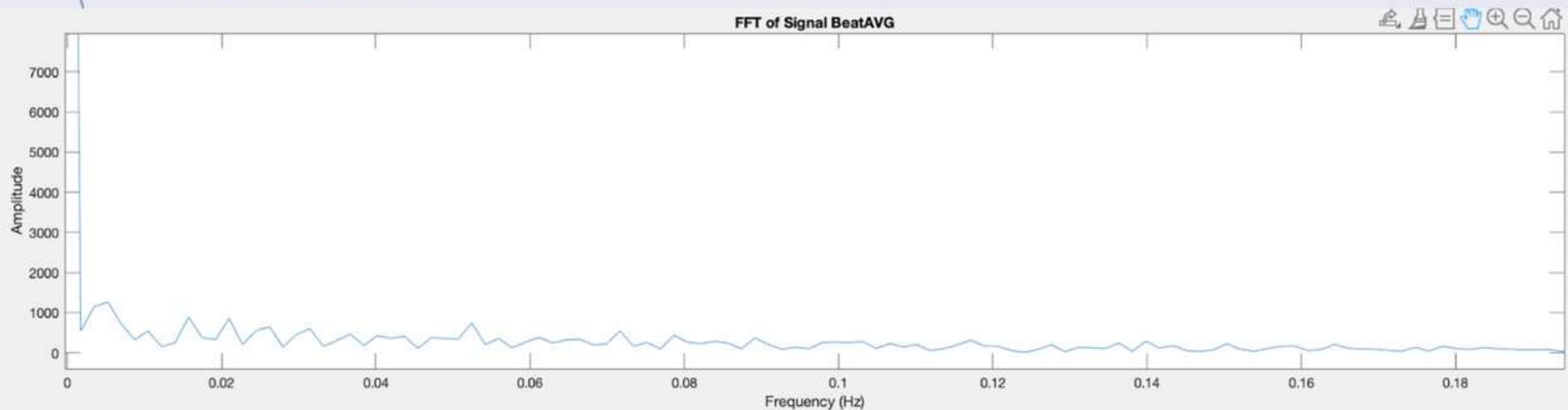
```
const int IR_THRESHOLD = 50000;
const int BPM_MIN = 40;
const int BPM_MAX = 150;
```

Data PreProcessing

Fourier
Transform

FIR Low-Pass
Filter

Convolution



```
% Sampling frequency in Hz (1 Hz in your case)  
fs = 1;
```

```
% FFT calculation  
fft_result = fft(BeatAVG_data);
```

Data PreProcessing

Fourier
Transform

FIR Low-Pass
Filter

Convolution

fir_coefficients.txt

```
6.2264927e-19 1.2014264e-03 2.7843278e-03 4.2273161e-03 3.94276  
-8.2567773e-03 -1.8583210e-02 -2.5389820e-02 -2.1235309e-02 5.9929992  
9.1888877e-02 1.4509908e-01 1.8490288e-01 1.9966067e-01 1.8490288e  
9.1888877e-02 3.9588113e-02 5.9929992e-18 -2.1235309e-02 -2.5389820e  
-8.2567773e-03 -2.4127659e-18 3.9427634e-03 4.2273161e-03 2.7843278  
6.2264927e-19
```

```
% Define a low-pass FIR filter  
cutoff_frequency = 0.1; % Cutoff frequency in Hz  
filter_order = 30; % Filter order  
b = fir1(filter_order, cutoff_frequency/(fs/2), 'low'); % Create the FIR filter
```

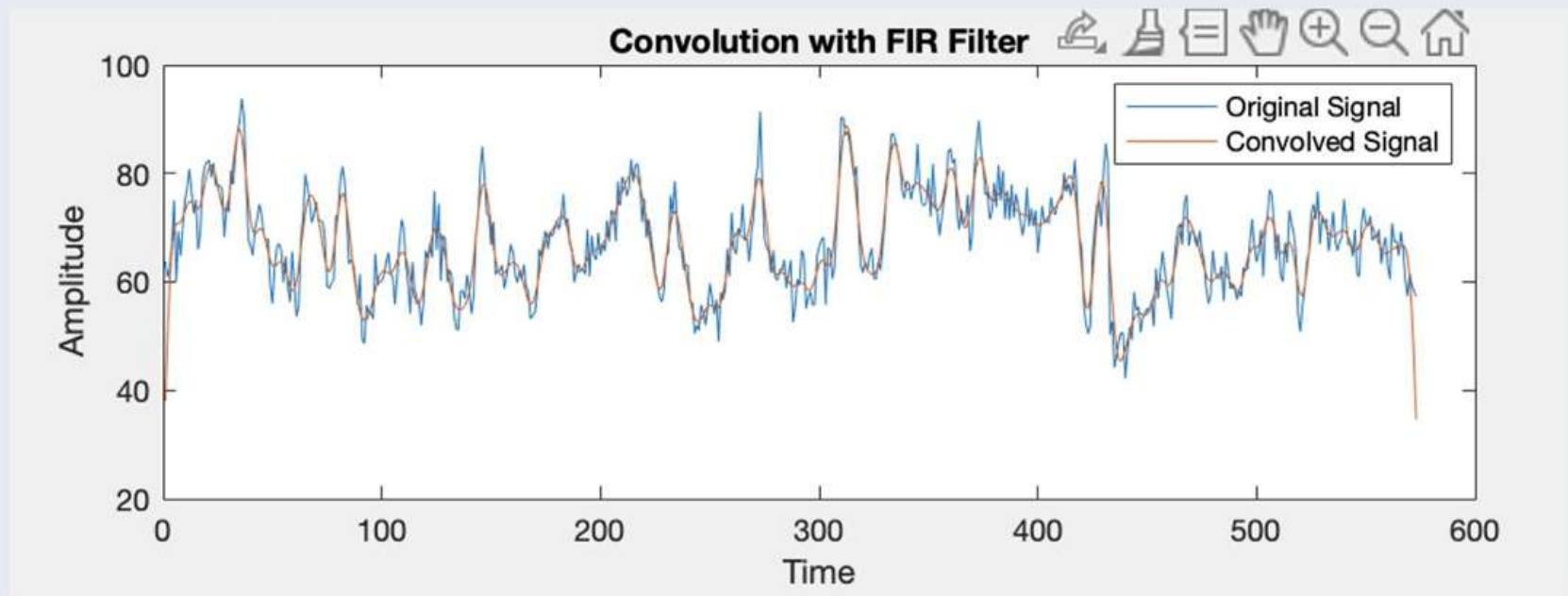
```
double firCoefficients[] = {fir_coefficients.txt};  
  
double shiftRegister[31];  
double filterOutput;  
  
double filterSample(double input) {  
    // Update the shift register  
    for (int i = 30; i > 0; i--) {  
        shiftRegister[i] = shiftRegister[i - 1];  
    }  
    shiftRegister[0] = input;  
  
    // Compute the filter output  
    filterOutput = 0;  
    for (int i = 0; i < 31; i++) {  
        filterOutput += firCoefficients[i] * shiftRegister[i];  
    }  
  
    return filterOutput;  
}
```

Data PreProcessing

Fourier
Transform

FIR Low-Pass
Filter

Convolution



Data Segmentation

In Arduino

```
void analyzeBPM(int bpm, unsigned long elapsedTime) {
    if (bpm > 73) {
        wakeTimes.push_back(elapsedTime);
    } else if (bpm < 45) {
        deepSleepTimes.push_back(elapsedTime);
    } else if (bpm > 45 && bpm < 58) {
        lightSleepTimes.push_back(elapsedTime);
    } else if (bpm > 58 && bpm < 73) {
        remSleepTimes.push_back(elapsedTime);
    }
}
```

In Matlab

```
Serial.println("Point le plus haut : " + String(highestBPM));
Serial.println("Point le plus bas : " + String(lowestBPM));

// Calculer et afficher la moyenne des BPM
float averageBPM = calculateAverageBPM(allBPMs);
Serial.println("Moyenne des BPM : " + String(averageBPM));

// Calculer et afficher le temps total passé hors de la phase WAKE
unsigned long totalTimeNotInWake = calculateTotalTimeNotInWake(allTimes, wakeTimes);
Serial.println("Temps total passé hors de la phase WAKE : " + formatTime(totalTimeNotInWake));

Serial.println("Temps total passé en WAKE (BPM > 73): " + formatTime(calculateTotalTime(wakeTimes)));
Serial.println("Temps total passé en REM_SLEEP (58 < BPM < 73): " + formatTime(calculateTotalTime(remSleepTimes)));
Serial.println("Temps total passé en LIGHT_SLEEP (45 < BPM < 58): " + formatTime(calculateTotalTime(lightSleepTimes)));
Serial.println("Temps total passé en DEEP_SLEEP (BPM < 45): " + formatTime(calculateTotalTime(deepSleepTimes)));
```

```
unsigned long calculateCurrentTime(const String& line) {
    int commaIndex = line.indexOf(',');
    String timeString = line.substring(0, commaIndex);
    int hours = timeString.substring(0, timeString.indexOf(':')).toInt();
    int minutes = timeString.substring(timeString.indexOf(':') + 1, timeString.lastIndexOf(':')).toInt();
    int seconds = timeString.substring(timeString.lastIndexOf(':') + 1).toInt();
    return (hours * 3600 + minutes * 60 + seconds) * 1000;
}
```


Data Segmentation

In Arduino

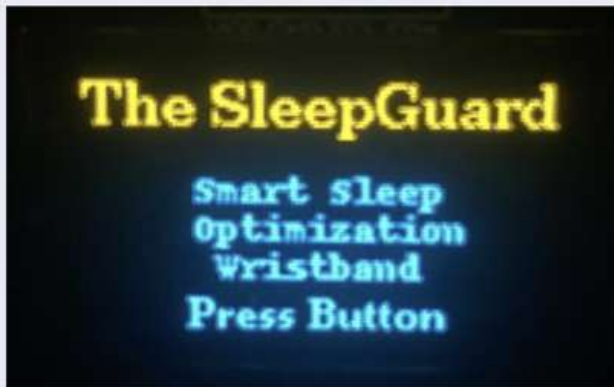
```
1 % Charge the data
2 filename = 'C:\Users\adele\Documents\Arduino\heart_rate_dataprof.csv';
3 opts = detectImportOptions(filename, 'NumHeaderLines', 0);
4 opts = setvartype(opts, {'char', 'double'});
5 data = readtable(filename, opts);
6 data.Properties.VariableNames = {'Time', 'BPM'};
7
8 % Converting strings to datetime
9 try
10     data.Time = datetime(data.Time, 'InputFormat', 'HH:mm:ss');
11 catch
12     data.Time = datetime(data.Time);
13 end
14
15 % Percent Thresholds for Different Sleep Phases
16 WAKE_THRESHOLD = 85;
17 REM_SLEEP_THRESHOLD = 70;
18 LIGHT_SLEEP_THRESHOLD = 65;
19 DEEP_SLEEP_THRESHOLD = 55;
20
21 % Identify the sleep phases
22 sleep_phase = repmat("LIGHT_SLEEP", height(data), 1);
23 sleep_phase(data.BPM > WAKE_THRESHOLD) = "WAKE";
24 sleep_phase(data.BPM < DEEP_SLEEP_THRESHOLD) = "DEEP_SLEEP";
25 sleep_phase(data.BPM > LIGHT_SLEEP_THRESHOLD & data.BPM <= WAKE_THRESHOLD) = "LIGHT_SLEEP";
26 sleep_phase(data.BPM > DEEP_SLEEP_THRESHOLD & data.BPM <= REM_SLEEP_THRESHOLD) = "REM_SLEEP";
```

In Matlab

```
28 % Initialization of Time Counters for Each Phase
29 tempsWake = 0;
30 tempsDeepSleep = 0;
31 tempsRemSleep = 0;
32 tempsLightSleep = 0;
33
34 % Calculate Time Spent in Each Phase
35 for i = 2:height(data)
36     deltaT = seconds(data.Time(i) - data.Time(i-1));
37     switch sleep_phase(i)
38         case "WAKE"
39             tempsWake = tempsWake + deltaT;
40         case "DEEP_SLEEP"
41             tempsDeepSleep = tempsDeepSleep + deltaT;
42         case "REM_SLEEP"
43             tempsRemSleep = tempsRemSleep + deltaT;
44         case "LIGHT_SLEEP"
45             tempsLightSleep = tempsLightSleep + deltaT;
46     end
47 end
48
49 % Convert Durations to Hours:Minutes:Seconds
50 tempsWake = duration(0, 0, tempsWake, 'Format', 'hh:mm:ss');
51 tempsDeepSleep = duration(0, 0, tempsDeepSleep, 'Format', 'hh:mm:ss');
52 tempsRemSleep = duration(0, 0, tempsRemSleep, 'Format', 'hh:mm:ss');
53 tempsLightSleep = duration(0, 0, tempsLightSleep, 'Format', 'hh:mm:ss');
```

Display & Reporting

The Main Page The Main Menu Sleep Measure Sleep Analysis



```
u8g2.begin();
u8g2.setFont(u8g2_font_ncenB10_tr);
u8g2.clearBuffer();
u8g2.drawStr(1, 12, "The SleepGuard");

u8g2.setFont(u8g2_font_courB08_tr);
u8g2.drawStr(30, 30, "Smart Sleep ");

u8g2.setFont(u8g2_font_courB08_tr);
u8g2.drawStr(30, 40, "Optimization");

u8g2.setFont(u8g2_font_courB08_tr);
u8g2.drawStr(30, 50, " Wristband");

u8g2.setFont(u8g2_font_ncenB08_tr);
u8g2.drawStr(25, 63, " Press Button");

u8g2.sendBuffer();
```

Display & Reporting

The Main Page **The Main Menu** Sleep Measure Sleep Analysis



```
u8g2.setFont(u8g2_font_ncenB14_tr);
u8g2.clearBuffer();
u8g2.drawStr(30, 15, "MENU");

u8g2.setFont(u8g2_font_ncenB08_tr);

if (menuOption == 0) {
    u8g2.drawStr(10, 35, "> Starting Sleep ");
    u8g2.drawStr(10, 55, " Starting Analysis");
} else {
    u8g2.drawStr(10, 35, " Starting Sleep ");
    u8g2.drawStr(10, 55, "> Starting Analysis");
}

u8g2.sendBuffer();
```

Display & Reporting

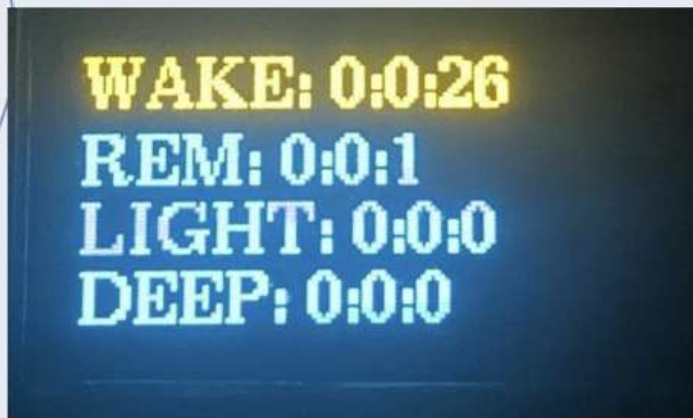
The Main Page The Main Menu **Sleep Measure** Sleep Analysis



```
if (buttonState == LOW && !buttonPressed) {  
  if ((millis() - pressStartTime) > longPressDuration) {  
    u8g2.setFont(u8g2_font_ncenB14_tr);  
    u8g2.clearBuffer();  
  
    if (menuOption == 1) {  
      SD.remove("heart_rate_data.csv");  
      u8g2.setFont(u8g2_font_ncenB10_tr);  
      u8g2.clearBuffer();  
      u8g2.drawStr(1, 45, "RECORDING... ");  
      u8g2.sendBuffer();  
      delay(1000);  
      while (stopLoop == false) {  
        loopMeasure();  
      }  
    }  
  }  
}
```


Display & Reporting

The Main Page The Main Menu Sleep Measure **Sleep Analysis**



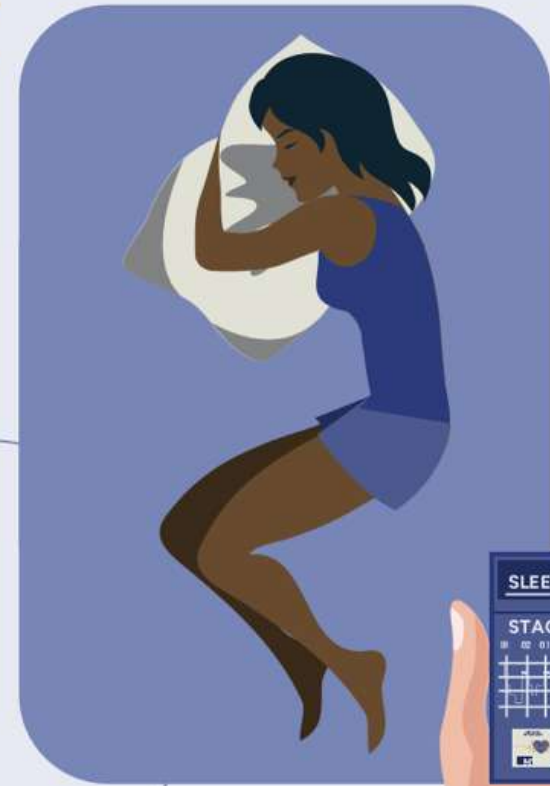
```
} else {  
    u8g2.setFont(u8g2_font_ncenB10_tr);  
    u8g2.clearBuffer();  
    analyzeSleep();  
    u8g2.setCursor(0, 15);  
    u8g2.print("WAKE: ");  
    u8g2.print(formatTime(calculateTotalTime(wakeTimes)));  
  
    u8g2.setCursor(0, 30);  
    u8g2.print("REM: ");  
    u8g2.print(formatTime(calculateTotalTime(remSleepTimes)));  
  
    u8g2.setCursor(0, 45);  
    u8g2.print("LIGHT: ");  
    u8g2.print(formatTime(calculateTotalTime(lightSleepTimes)));  
  
    u8g2.setCursor(0, 60);  
    u8g2.print("DEEP: ");  
    u8g2.print(formatTime(calculateTotalTime(deepSleepTimes)));  
}  
  
u8g2.sendBuffer();
```

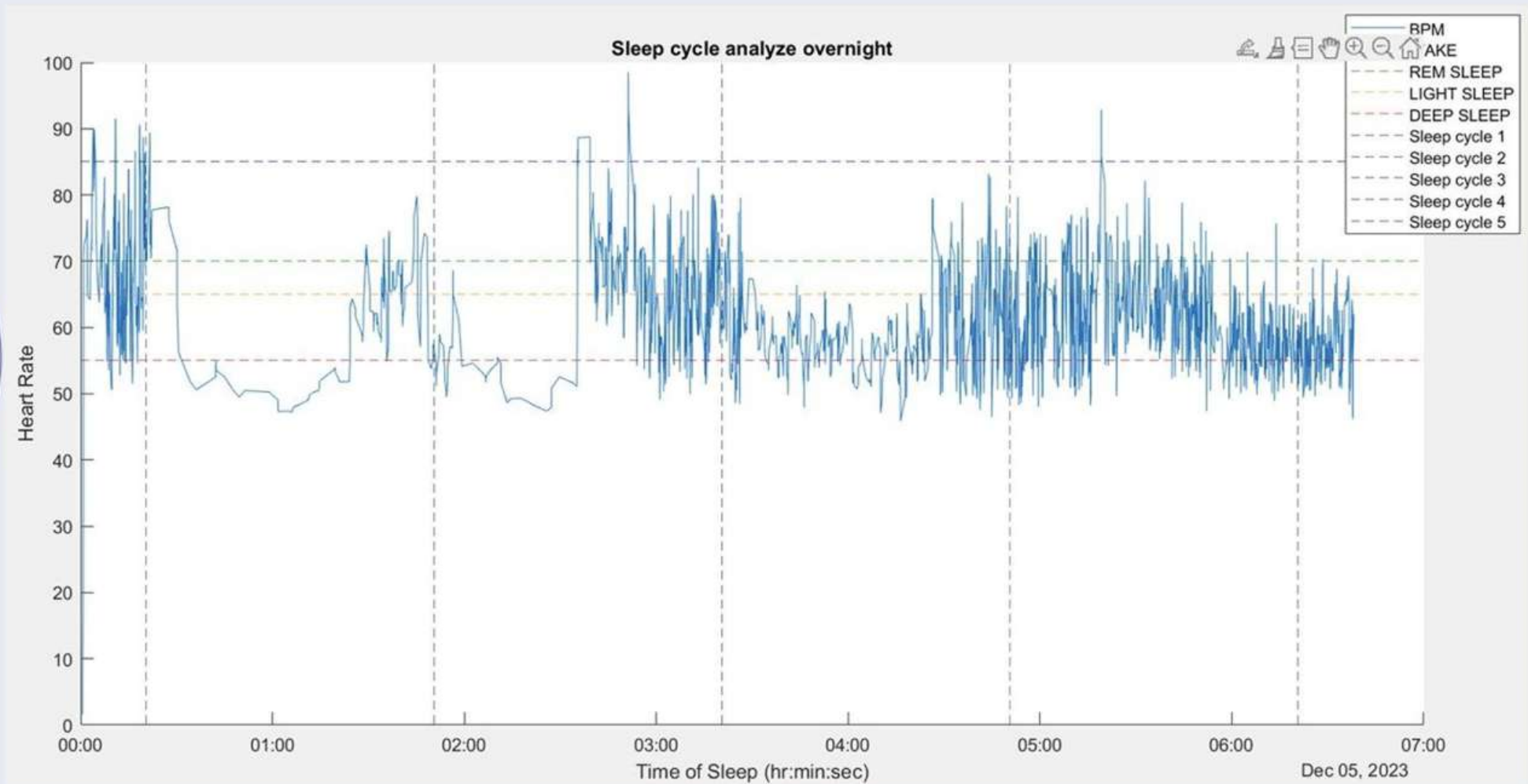

Modeling & Interpretation

```
80 % Add Horizontal Lines for Each Threshold
81 yline(WAKE_THRESHOLD, '--', 'Color', [0, 0, 0.5], 'DisplayName', 'WAKE'); % Bleu marine
82 yline(REM_SLEEP_THRESHOLD, '--', 'Color', [0, 0.5, 0], 'DisplayName', 'REM SLEEP'); % Vert foncé
83 yline(LIGHT_SLEEP_THRESHOLD, '--', 'Color', [1, 0.5, 0], 'DisplayName', 'LIGHT SLEEP'); % Orange
84 yline(DEEP_SLEEP_THRESHOLD, 'r--', 'DisplayName', 'DEEP SLEEP'); % Rouge
85
86 % Calculate and Plot Vertical Lines for Each New 90-Minute Cycle
87 cycle_duration = 1.5; % 1 sleep cycle in hours
88 timestamps = data.Time; % Use Timestamps from the Data Table
89 start_time = timestamps(1) + minutes(20); % start 20 min after the start of the recording
90 end_time = timestamps(end);
91 num_cycles = floor((end_time - start_time) / hours(cycle_duration));
92
93 for i = 0:num_cycles
94     cycle_start = start_time + hours(i * cycle_duration);
95     xline(cycle_start, '--', 'DisplayName', ['Sleep cycle ' num2str(i+1)]);
96 end
97
98 hold off;
99 xlabel('Time of Sleep (hr:min:sec)');
100 ylabel('Heart Rate');
101 title('Sleep cycle analyze overnight');
102 legend;
```

04

Results





05

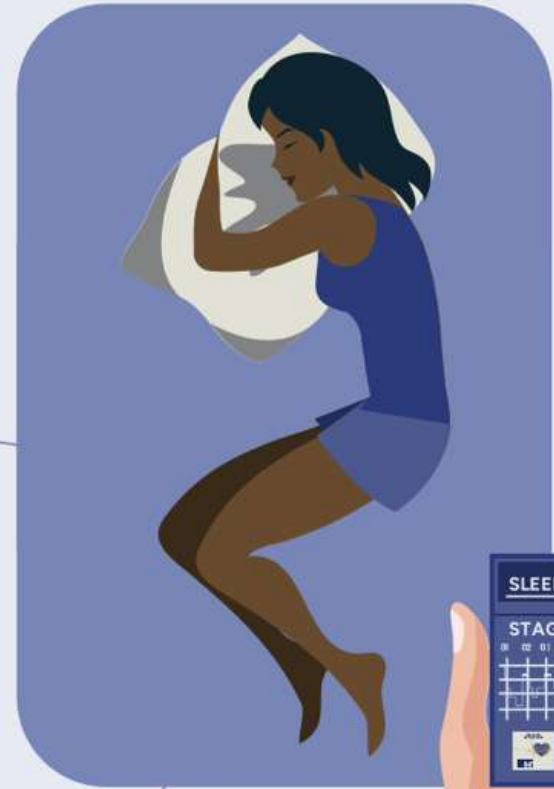
Conclusion





06

Bibliography



Resources

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